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Study of enzyme sensors with wide, adjustable measurement ranges for in-situ monitoring of biotechnological processes

Stefan Mross[a,*], Tom Zimmermann[a], Sebastian Zenzes[b], Michael Kraft[c], Holger Vogt[a]

[a] Fraunhofer Institute for Microelectronic Circuits and Systems, Finkenstrasse 61, 47057 Duisburg, Germany
[b] University of Applied Sciences Niederrhein, Krefeld, Germany
[c] Montefiore Institute, Department of Electrical Engineering and Computer Science, University of Liège, Belgium

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ABSTRACT

We report on a study of enzymatic glucose and lactate sensors for measurements of wide concentration ranges from 1 mM up to 600 mM and 900 mM, respectively, in biotechnological processes. Diffusion-limiting polyurethane membranes were used to extend the linear measurement range and the influence of two fabrication parameters on membrane properties and sensor performance evaluated. The polymer concentration was varied between 4% and 10% and the ratio of the solvents tetrahydrofuran and dimethylformamide between 9:1 and 1:9. Surface morphology and permeability of the membranes and the linear measurement range of membrane-covered enzyme sensors were determined. A distinct relationship between the different results was found. The sensors retained their functionality after being sterilized using gamma and electron beam irradiation. The long-term performance of the sensors was evaluated and good performance in the glucose monitoring of a culture of *Saccharomyces cerevisiae* over 13 days was shown.

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1. Introduction

The demand for products with high expression rates, including monoclonal antibodies, therapeutic proteins and vaccines, is rapidly growing [1,2]. As a consequence, biotechnological processes in production and research become increasingly complex and sophisticated. At the same time, high efficiency and product quality are key requirements for various processes, be they new or well-established [3]. The monitoring of glucose and lactate is of major importance for good process control, high product quality and maximum yield. The precise control of glucose allows increasing antibody yield, reducing applied nutrients, ensuring reproducible glycosylation and preventing stuck fermentations [4]. Similarly, lactate is one of the main metabolites in mammalian cell cultures [5] and gives information about the health of the cells and their metabolic activity [3].

Today, glucose and lactate in bioreactors are typically measured after sampling in external electrochemical analyzers [3]. However, sampling requires additional ports, bears the risk of contamination and delays the measurement results. In contrast, continuous in-situ glucose and lactate measurements could provide real-time data and thus be of great benefit for many applications.

When designing in-situ sensors for glucose and lactate, the wide concentration ranges of these analytes in bioprocesses have to be considered. Concentrations go up to around 50 mM for mammalian cell cultures but can reach values up to 1500 mM glucose in alcoholic fermentation [6] and 1000 mM lactate in the production of biodegradable polymers [7]. Sensors for such wide concentration ranges have rarely been reported since most of the research in glucose and lactate sensing is directed towards applications in health care. Demonstrated sensors for in-situ measurements in bioreactors are also limited to concentration ranges up to 50 mM [8,9].

Diffusion-limiting membranes are a way to extend the linear range of enzyme sensors [8–12]. Since most of the research in this field is limited to low concentrations, membranes have been optimized for this range, and there are only few studies aimed at higher ranges [13–15]. In this contribution, we develop glucose and lactate sensors with polyurethane (PU) membranes for wide concentration ranges. PU has often been used as a material for diffusion-limiting membranes for enzyme sensors [11,16,17], however, the influence of fabrication parameters such as polymer concentration and solvent composition on the sensor performance has rarely been studied. These parameters have, however, been found to have significant impact on the membrane structure in the context of tissue engineering [18]. Hence, we decided to evaluate the influence of polymer concentration and solvent composition on membrane morphology, permeability and the linear range of fabricated enzyme sensors.

* Corresponding author.
E-mail address: stefan.mross@ims.fraunhofer.de (S. Mross).

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2. Experimental method

The glucose and lactate sensors were based on enzymatic conversion of the target molecules using glucose oxidase or lactate oxidase, followed by oxidation of the generated H_2O_2 and amperometric detection of the current by a potentiostat [4]. Enzymatic reactions can be described by the Michaelis-Menten model, in which the general form of an enzymatic reaction is written as [19]

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \overset{k_2}{\rightarrow} E + P, \quad (1)$$

where E denotes the enzyme, S its substrate, P the product of the enzymatic reaction and k_1, k_{-1} and k_2 are the rate constants for the individual steps of the reaction.

A helpful quantity in the description of enzyme kinetics is the Michaelis constant K_M. It can be seen as a measure for the affinity of the enzyme for its substrate. The lower K_M, the higher the affinity. Under the assumption that the product formation is much faster than dissociation of the enzyme-substrate complex, i.e. $k_2 \ll k_{-1}$, K_M is calculated as

$$K_M \approx \frac{k_{-1}}{k_1}. \quad (2)$$

Up to K_M the relationship between the substrate concentration and the velocity of the enzymatic reaction can be considered linear. The Michaelis constants for glucose oxidase and lactate oxidase are 33 mM and 0.7 mM, respectively [20].

For glucose oxidase in particular, the enzymatic reaction mechanism is as follows [21]:

$$\beta\text{-d-Glucose} + \text{GOx-FAD} \rightarrow \text{GOx-FADH}_2 + \delta\text{-d-gluconolactone} \quad (3)$$

$$\text{GOx-FADH}_2 + O_2 \rightarrow \text{GOx-FAD} + H_2O_2. \quad (4)$$

In electrochemical measurements typically a three-electrode setup is employed, which includes a working electrode (WE) and a counter electrode (CE) made from noble metals, and a silver/silver chloride reference electrode (RE). A potentiostat keeps WE at a constant potential in respect to RE. The H_2O_2 generated in the enzymatic reaction is oxidized at the working electrode, yielding two electrons:

$$H_2O_2 \rightarrow 2e^- + O_2 + 2H^+. \quad (5)$$

The generated current is then detected by the potentiostat.

To achieve wide measurement ranges in enzyme sensors, diffusion-limiting membranes can be fabricated on top of the enzyme layer. For such sensors the upper limit S of the linear measurement range can be expressed as [23]

$$S = 1 + 0.5 k_a l_m / D_{S,m}. \quad (6)$$

In Eq. (6), k_a includes several parameters of the enzyme layer, such as the Michaelis constant, enzyme concentration and enzyme layer thickness [22]. $D_{S,m}$ is the diffusion coefficient of substrate in the membrane and l_m is the membrane thickness. Since the response time of the sensor rises with the square of l_m but scales only linearly with the inverse of $D_{S,m}$ [23], we focused on adjusting the diffusion coefficient to extend the linear measurement range in this study.

The enzyme electrodes were designed with concentric working (WE), counter (CE) and reference (RE) electrodes. Thanks to the concentric arrangement an even distribution of the electric field lines between the electrodes was ensured and the sensor was made independent of the direction of flow in stirred solutions in comparison to finger shaped electrodes. The WE had a diameter of 400 μm, CE and RE had a width of 100 μm and the spacing was 50 μm, giving a total diameter of 1 mm. This diameter matched the droplet size of 0.1 μl of enzyme solution that was used during the functionalization of the electrodes. The electrodes were fabricated on oxidized silicon substrates by evaporation of gold or platinum with titanium as an adhesion layer, followed by a lift-off process. The design of the sensor chip is shown in Fig. 1. Together with the enzyme sensors other sensor elements were integrated, forming a multi-sensor system for the monitoring of biotechnological processes. A description of the other sensors can be found elsewhere [24,25].

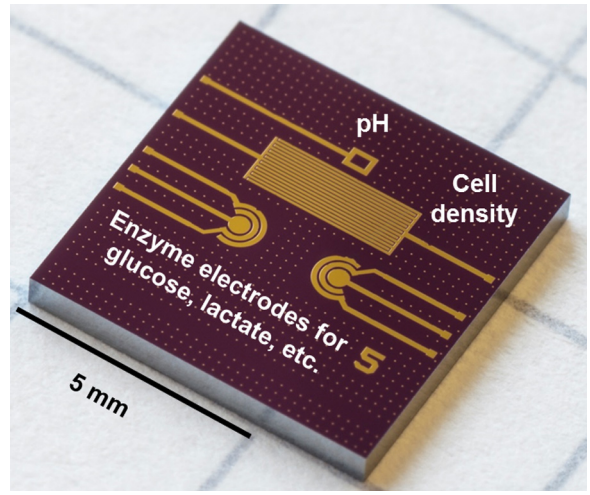

Fig. 1. Photograph of silicon multi-sensor chip with concentric three-electrode configurations for amperometric enzyme sensors; chip size was 7.16 mm × 7.16 mm; a description of the other sensors can be found elsewhere [24,25].

To functionalize the electrodes a solution of 10 mg glucose oxidase (GOx) or lactate oxidase (LOx), 10 mg bovine serum albumin and 25 μl glutaraldehyde was prepared in 500 μl of 10 mM phosphate buffered saline (PBS). 0.1 μl of the solution were dispensed on the three-electrode configuration and allowed to dry in air. The diffusion-limiting membrane was made from polyurethane (Selectophore®, Sigma Aldrich, USA), which is biocompatible and permeable for oxygen, glucose and lactate. In order to evaluate the influence of polymer concentration and solvent composition on the linear range, a series of membranes was produced. PU solutions in concentrations of 4%, 6%, 8% and 10% (w/w) were prepared with a mixture of the solvents tetrahydrofuran (THF) and dimethylformamide (DMF). Higher concentrations were too viscous to be dispensed using microliter pipettes. The THF-DMF ratio was adjusted to 9:1, 7:3, 5:5, 3:7 or 1:9, respectively, giving five series of solutions. The PU used in this study was found to be indissoluble in the mixture THF:DMF 1:9 even after several days, so this series was not employed in the further experiments. 0.15 μl of the PU solution were deposited on the enzyme layer and dried in air.

The diffusion coefficient for glucose in the membranes was determined using a custom-made diffusion cell connected to a stirred reservoir containing a 2 M glucose solution and to a second stirred reservoir containing deionized water. Samples were drawn regularly from the reservoirs, the glucose concentrations measured and the diffusion coefficient D determined according to the formula [26]

$$D = \frac{1}{\beta t} \ln\left(\frac{c_I^0 - c_{II}^0}{c_I - c_{II}}\right), \quad (7)$$

with

$$\beta = \frac{A}{l}\left(\frac{1}{V_I} + \frac{1}{V_{II}}\right). \quad (8)$$

In Eqs. (7) and (8), c_I and c_{II} are the glucose concentrations in the first and second reservoir at time t, $c_I{}^0$ and $c_{II}{}^0$ are the respective glucose

concentrations at time $t = 0$, A and l are the area and thickness of the membrane and V_I and V_{II} are the volumes of the two reservoirs.

To study the surface morphology of the membranes, PU solutions were dispensed on oxidized silicon substrates and observed under a scanning electron microscope (SEM, 7 kV accelerating voltage, magnification 50× up to 70×) and an atomic force microscope (AFM, tapping mode, cantilever spring constant: 1.3–3.4 N/m).

The thickness of the PU membranes was determined for a THF-DMF ratio of 5:5 by dispensing PU solutions of different concentrations onto an oxidized silicon substrate and examining the cross section in the SEM. The thickness ranged from 1.5 μm for membranes with 4% PU to 2.5 μm for membranes with 10% PU.

Before amperometric measurements electrodes were conditioned in PBS for 12 h. Glucose solutions were prepared in PBS. Measurements were performed at room temperature using a Multi-EmStat3 potentiostat (PalmSens, The Netherlands). Due to concerns about the stability of the on-chip reference electrode an external reference electrode (Micro-Dri-Ref 450, WPI, USA) was used during the experiments. In future work the integration of a screen printed Ag/AgCl reference electrode on the chip is planned. The potential between the working and reference electrode was set to 700 mV, ensuring oxidation of the H_2O_2 generated in the enzymatic reactions.

3. Results

3.1. Membrane morphology

To evaluate the influence of varying PU concentrations and solvent compositions on the membrane morphology, PU membranes were examined under an SEM and an AFM. First, the influence of the PU concentration was evaluated using a fixed solvent composition of THF:DMF 9:1. Such a high THF content has been used by others [11,16]. As can be seen from the first row in Fig. 2a, the surface of a membrane with 4% PU contained pores with diameters from about 30 μm to 100 μm and some randomly distributed larger pores. Most of the pores did not run through the whole membrane thickness, suggesting that the membrane was predominantly made up of blind pores and hence should be regarded as foam.

With increasing PU concentration number and depth of the pores decreased, leading to areas with higher material density. The AFM images shown in the upper row of Fig. 2b confirmed that with increasing polymer concentration also on a smaller scale of 90 μm × 90 μm a finer surface structure was achieved. The observed morphological characteristics were expected to distinctly influence the membrane permeability for analytes such as glucose and lactate since a denser material with finer pores was expected to resist diffusion more than a highly porous material.

As a second parameter, the ratio of the solvents tetrahydrofuran and dimethylformamide was varied as THF:DMF 9:1, 7:3, 5:5, 3:7, using a fixed PU concentration of 6%. The SEM images in the second row of Fig. 2a indicate that with the solvent ratio changing from 9:1 to 5:5, the pore density declined, fewer pores formed, and a more homogeneous surface was observed. The same trend could be found in the AFM pictures in the lower row of Fig. 2b, which showed a finer morphology for an increasing DMF ratio. The ratio THF:DMF 3:7 lead to a rather dense surface, which, however, contained big voids and defects. These were ascribed to a lower solubility of the used PU in DMF than in THF: conceivably, the polymer precipitated in the DMF, already formed pores collapsed, and a dense membrane formed instead of a porous one [27]. This phenomenon is well-known in phase inversion processes for membrane fabrication [28] and is proposed here as an alternative explanation for the observed relationship between solvent composition and membrane properties. An alternative explanation for the observed membrane morphology is the difference in the vapor pressures of THF and DMF, which is described in subsection 3.2.

3.2. Membrane permeability

To assess the influence of PU concentration and solvent composition on the permeability of membranes and confirm the observations from the microscope pictures, diffusion coefficients for glucose were measured for PU concentrations from 4% to 10%, and the THF-DMF ratio was exemplarily chosen as 9:1. The results depicted in Fig. 3a confirmed the anticipated influence of the observed surface morphology on membrane permeability. With increasing PU concentration the diffusion coefficient declined from 4.7×10^{-11} m^2/s to 9.7×10^{-12} m^2/s with a square relationship.

A similar relationship could be found for the THF-DMF ratio varying between 9:1 and 5:5 at a fixed PU concentration of 6%, as depicted in Fig. 3b. Here, the diffusion coefficients ranged from 2.6×10^{-11} m^2/s to 1.51×10^{-11} m^2/s. A further increase of the DMF amount led to an increased diffusion coefficient of 3.21×10^{-11} m^2/s, which had been expected because of the observed voids in the membrane at this solvent ratio. These results pointed to the fact that both an increased PU concentration and a higher amount of DMF up to a ratio of 5:5 lead to a membrane that was less permeable to glucose and thus was expected to increase the measurement range of enzyme sensors. The influence of the THF-DMF ratio was likely to be due to the different vapor pressures of the solvent compositions. The vapor pressures of DMF and THF are 0.49 kPa and 21.6 kPa at 25 °C, respectively [29], meaning that DMF evaporates slower than THF. Hence, the more DMF was contained in the solvent mixture, the slower the solvent evaporated, leading to a more homogeneous membrane surface with less pores and a lower permeability.

3.3. Calibration measurements

After the morphology and permeability of the polyurethane membranes had been evaluated, the linear measurement range of enzyme sensors coated with these membranes was studied. These experiments were first carried out for glucose sensors, and the found results then applied to lactate sensors. Repeated measurements (n = 5) were performed for each sensor and concentration.

As a first step the influence of the solvent composition was investigated at a fixed PU concentration, which was exemplarily chosen as 6%. From the results depicted in Fig. 4a a parallel could be drawn between the observed membrane morphology, permeability and the linear range. With increasing DMF concentration the linear limit rose from 200 mM for a solvent ratio of THF:DMF 9:1 to 400 mM for a ratio of THF:DMF 5:5 while the sensitivity decreased. The sensitivity was taken as the slope of the regression line in the linear range and determined to be between 0.89 nA/mM at THF:DMF 9:1 and 0.27 nA/mM at THF:DMF 5:5. As anticipated, the ratio THF:DMF 3:7 gave the smallest extension to 50 mM and the highest sensitivity, which could be ascribed to the large voids found in the membrane, which did not limit the flux of glucose through the membrane.

As the next step, the influence of the PU concentration was studied at a fixed solvent ratio of THF:DMF 9:1 for glucose and lactate sensors. In Fig. 4b and c the calibration curves of glucose and lactate sensors with these PU membranes are shown. With increasing PU concentration an increase in the linear measurement range was observed for both sensor types, leading to ranges from 200 mM to 300 mM for glucose and from 300 mM to 600 mM for lactate. Sensitivities decreased as well from 1.82 nA/mM at PU 4% to 0.21 nA/mM at PU 10% for glucose and from 1.56 nA/mM to 0.17 nA/mM for lactate. Again, these findings correlated with the results of the studies

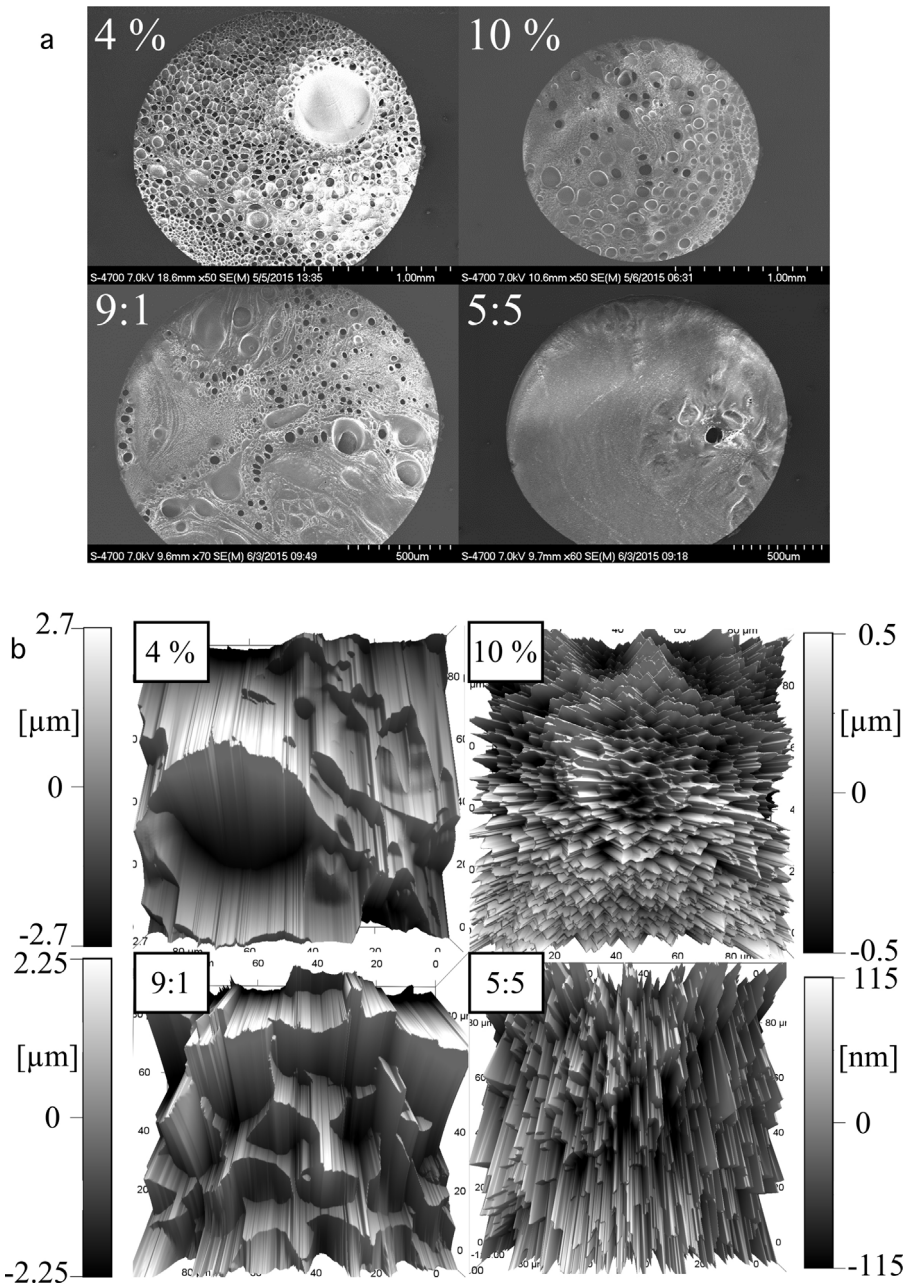

Fig. 2. Surface morphology of polyurethane membranes with polymer concentrations of 4% and 10% (THF:DMF 9:1) and solvent compositions of THF:DMF 9:1 and 5:5 (PU 6%), recorded with (a) a scanning electron microscope (SEM) and (b) an atomic force microscope (AFM), AFM scan size 90 µm × 90 µm.

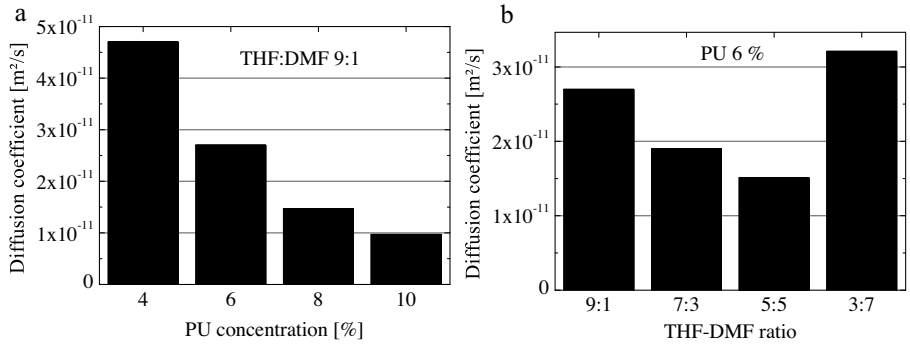

Fig. 3. Diffusion coefficients of PU membranes for glucose with varying (a) PU concentrations and (b) THF-DMF ratios.

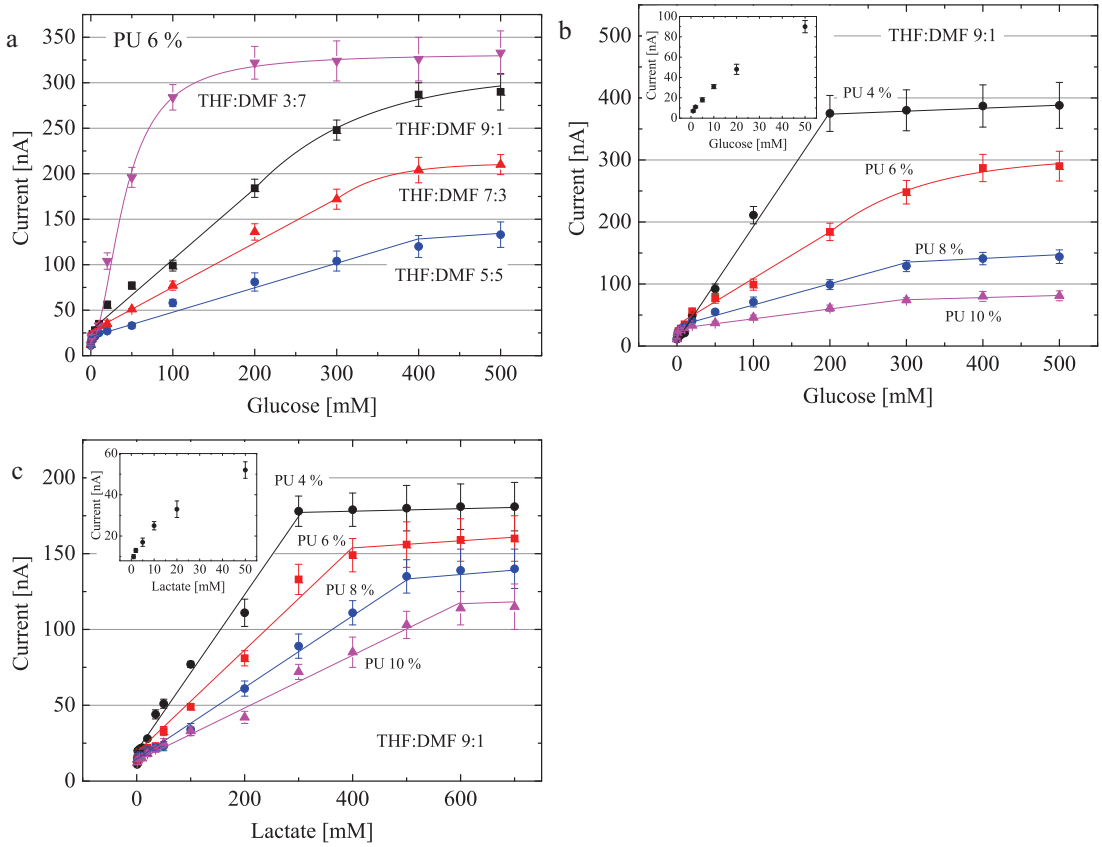

Fig. 4. Calibration curves of (a) glucose sensors with different solvent mixtures at an exemplary PU concentration of 6%; (b) glucose sensors with different PU concentrations, ratio THF:DMF 9:1; (c) lactate sensors with different PU concentrations, ratio THF:DMF 9:1; arrows mark linear range, insets show calibration curves of sensors with PU 4% between 1 mM and 50 mM; error bars show standard deviations of repeated measurements (n = 5).

on surface morphology and permeability of the PU membranes. The fact that polymer concentrations above 6% or 8% did not extend the linear range further could be attributed to a non-sufficient oxygen supply to the enzyme layer under the membrane [9,10].

The calibration measurements also allowed identifying the sensors with the highest sensitivity and widest range in this study. For glucose the highest sensitivity was achieved with a solvent ratio of THF:DMF 9:1 and a PU concentration of 4%, and the widest linear range with a THF-DMF ratio of 5:5 and a PU concentration of 6%. For lactate the highest sensitivity was achieved at a THF-DMF ratio of 9:1 and a PU concentration of 4%, and the widest linear range at the same solvent ratio and a polymer concentration of 10%.

For determination of the resolution it was stated that the resolution cannot be smaller than the standard deviation of the sensors. The standard deviation of the sensors was assessed at a low concentration of 10 mM and a high concentration that was chosen as the upper limit of the respective sensor. From the regression analysis the obtained current standard deviations were converted into those for concentration. Then it was ascertained that the sensors give a distinguishable reading when a concentration of glucose or lactate equal to the concentration standard deviation was added to the measurement solution. In the low concentration range resolution was between 0.6 mM and 5.9 mM, depending on sensor sensitivity. For high concentrations the resolution was above 15 mM for all sensors except for the glucose sensor with THF-DMF ratio of 3:7, which had a resolution of 3.6 mM at a concentration of 50 mM. The resolution in the high concentration range is expected to greatly improve when manual functionalization of sensors is replaced by automated methods.

The membrane series with the solvent ratio of THF:DMF 5:5 was identified as the one giving the widest linear measurement range. Hence, the influence of the polymer concentration was studied at this ratio for glucose and lactate sensors, and the linear range for glucose and lactate was determined to extend up to 600 mM and 900 mM, respectively. For further information about sensors with these membrane parameters the reader is referred to [25].

The achieved measurement ranges for glucose and lactate allow application of the sensors in mammalian cell cultures. The sensors could, however, also be used for in-situ measurements in processes such as alcoholic fermentation [6] and the production of lactate for biodegradable polymers [7].

3.4. Sterilization

All equipment introduced into bioreactors needs to be sterilized. Physical sensors without biological elements are typically steam sterilized, however, for enzyme sensors, the high temperature above 120 °C of this sterilization procedure is prohibitive because of the irreversible denaturation of enzymes at high temperatures, leading to partial or complete loss of the biomolecules' functionality. As a solution, other sterilization procedures have been proposed, among them treatment with chemicals such as glutaraldehyde [30], and gamma or electron beam irradiation [31,32]. In comparison to chemical treatment, irradiation offers a more convenient and faster way to sterilize sensors. Since the proposed enzyme sensors are mainly intended for single use, fabricated devices could be sterilized at the production site and shipped in sterile bags. The end-user would then only have to remove the device in a sterile environment and install it in the already sterilized bioreactor. In the case of single-use reactors, the chip could be already installed in the reactor before shipment and the setup sterilized as a whole. This would provide the customer with a ready-

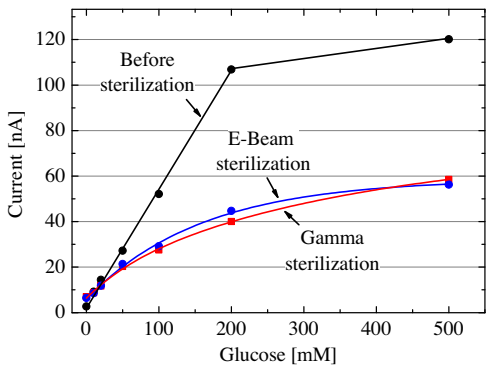

Fig. 5. Calibration curves of glucose sensors with PU membranes (PU 4%, THF:DMF 5:5) before and after gamma and electron beam sterilization at a dose of 25 kGy each.

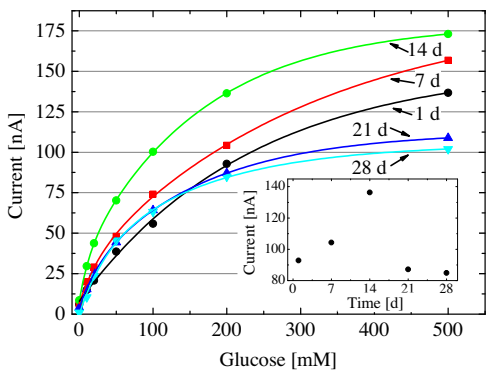

Fig. 6. Long-term performance of glucose sensors with PU membranes (PU 4%, THF:DMF 5:5) over 28 days; weekly calibration measurements and storage in PBS between measurements; inset shows sensor signal at 200 mM over time.

to-use product without the need for purchase, sterilization and installation of separate sensors.

To assess the general suitability of our functionalization method for sterilization, we sterilized glucose sensors with PU membranes. The parameters were chosen exemplarily as 4% PU and THF:DMF 5:5. Gamma and electron beam irradiation were chosen as the sterilization methods at a dose of 25 kGy each (mediscan GmbH, Austria). This dose has been shown to ensure a high sterility assurance level of 1×10^{-6} for many medical devices [33]. As can be seen from the calibration curves in Fig. 5, the sensors lost up to 60% of their sensitivity during the sterilization, and also the linear range was affected by the treatment. However, the sensors remained functional, showing promise for further developments using these sterilization methods. The impact of both sterilization methods on the sensitivity was almost similar, suggesting that both are equally suited for enzyme sensors. The long-term performance of sterilized sensors will be investigated in further experiments.

3.5. Long-Term performance

In-situ sensors for application in biotechnological processes have to perform continuously over the duration of the process, i.e. typically up to 30 days. Over this period of permanent contact with the cell culture medium changes of the sensor behavior are possible due to effects such as changing membrane permeability [4]. To investigate the long-term performance of glucose sensors with PU membranes, sensors with membranes (PU 4%, THF:DMF 5:5) were kept in 10 mM PBS, and calibration measurements were performed every 7 days. As can be seen from Fig. 6, the sensitivity of the sensors increased from 0.5 nA/mM at day 1 to 0.9 nA/mM at

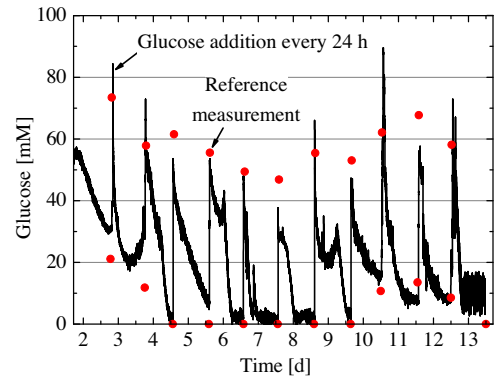

Fig. 7. Glucose monitoring in a culture of *Saccharomyces cerevisiae* over 13 days using a glucose sensor with a PU membrane (PU 6%, THF:DMF 5:5); glucose was added every 24 h to increase the concentration by 50 mM; curve: sensor signal, red dots: spectrophotometric reference measurements (Sigma Aldrich, USA). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

day 14. At the same time the linear measurement range decreased from 200 mM to 20 mM. During the following 14 days the sensitivity decreased to about the value at the first day. The behavior during the first 14 days could be ascribed to an increasing permeability of the diffusion-limiting membrane, originating from water uptake of the polymer or changes in the polymer structure [4]. In the second phase of the experiment, possibly, a decreasing enzyme activity acted contrarily to the increasing membrane permeability. For reliable long-term measurements the sensors can be re-calibrated by adding a bolus of known concentration to the reactor medium and determining the sensitivity from the induced signal increase.

3.6. Cell culture monitoring

To evaluate the suitability of the fabricated sensors for the monitoring of bioprocesses, the glucose level in a culture of *Saccharomyces cerevisiae* was monitored using a glucose sensor with a PU membrane with appropriate parameters for the intended application (PU 6%, THF:DMF 5:5). The culture was prepared in 10 mM PBS with an initial cell density of 1 g/l and a glucose concentration of 50 mM. The culture was continuously stirred and aerated at a rate of 150 l/h, and the pH was kept at pH 6 by addition of potassium hydroxide. Every 24 h glucose solution was added to increase the concentration by 50 mM, and every fourth day the culture was replaced to avoid the fermentation from coming to a halt. Samples were drawn immediately before and after the addition of glucose, diluted and analyzed using a spectrophotometric assay based on a colorimetric reaction with *o*-dianisidine (Sigma Aldrich, USA). The signal of the glucose sensor is shown in Fig. 7. During the first one and a half days the sensor was conditioned and calibrated. Consumption of glucose and replenishment can be clearly seen from the sensor signal, being in good correlation with the reference measurements. The first and third culture, running until day 4 and from day 8 onwards, respectively, did not consume all the glucose during a 24 h period. This can be seen from the sensor signal not reaching 0 mM. In general, the sensor signal showed good performance over the whole fermentation duration of 13 days, demonstrating a first application of the fabricated glucose sensors in bioprocess monitoring. Further measurements in different bioreactor media and at different reactor parameters (e.g. temperature and pH) are necessary to ensure full suitability of the developed sensors for application in bioprocesses.

4. Conclusion

We demonstrated enzyme sensors for glucose and lactate for in-situ application in biotechnological processes. Diffusion-limiting

polyurethane membranes were used and the sensor performance tuned by varying polymer concentration and solvent composition. The influence of these fabrication parameters was investigated in terms of membrane surface morphology, permeability and linear range. Using these membranes wide measurement ranges were achieved, covering a significant part of the concentrations found in bioprocesses. Thus, customer-specific sensors with adjustable measurement ranges can be fabricated for a broad range of biotechnological processes, ranging from mammalian cell cultures to fermentations and production of biodegradable polymers [6,7]. An application of the fabricated sensors in biotechnology was demonstrated in the glucose monitoring of a culture of *Saccharomyces cerevisiae* over 13 days and in long-term measurements over 28 days. The glucose sensors retained their functionality after sterilization by gamma and electron beam irradiation, meeting a critical requirement for application in bioreactors.

The sensor electrodes used in this work were fabricated on 8-inch silicon wafers in a CMOS-compatible fabrication process, and also the functionalization with enzymes and diffusion-limiting membranes can be carried out on this mass-fabrication scale. To obtain a fully-integrated, self-contained measurement system, the developed sensor elements could be fabricated on top of a CMOS substrate with readout circuits and interfaces to external control equipment. Such an easy-to-use, small and cost-effective system is expected to have great impact in the field of biotechnology.

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Biographies

S. Mross received the B.Sc. and M.Sc. degree in mechanical engineering from Ruhr University Bochum, Germany, in 2011 and 2012. He is currently pursuing his Ph.D. degree in electrical engineering at the Fraunhofer Institute for Microelectronic Circuits and Systems (IMS), Duisburg, Germany, working in the field of biosensors, multi sensor platforms and bioprocess monitoring. His research interests include biosensors, microsystems technology and control engineering.

T. Zimmermann heads the business unit Biohybrid Systems at the Fraunhofer Institute for Microelectronic Circuits and Systems (IMS), Duisburg, Germany. He graduated with a Dipl.-Ing. in solid-state electronics at the University Ulm, Germany, in 2002. Having received his Dr.-Ing. degree from the University Ulm, he joined as a post-doc the University of Notre Dame, IN, USA, in 2006, focusing on microfluidics, AlN/GaN and graphene HEMT devices. From 2009 to 2011, he was faculty member and Research Assistant Professor in nanotechnology and high-power terahertz devices. His research record: 3 filed US patents, 20+ peer-reviewed papers and 50+ conference proceedings.

S. Zenzes studied chemistry and biotechnology at the University of Applied Sciences Niederrhein, Germany, where he received his B.Sc. in 2013 and his M.Sc. in 2015. Since 2015 he has been pursuing his Ph.D. degree in the field of environmental analysis at the Institute for Biology and Chemistry, University of Hildesheim, Germany.

M. Kraft is a Professor of Micro- and Nanosystems at the University of Liege, Belgium. From 2012–2014, he was department head at the Fraunhofer Institute IMS in Duisburg, Germany, and W3 Professor of Integrated Micro- and Nanosystems at the University of Duisburg-Essen. From 1999–2012, he was a Professor of Micro-System-Technology at the University of Southampton, UK. He graduated with a Dipl.-Ing. (Univ.) in electrical and electronics engineering at the University Erlangen-Nürnberg in 1993, and was awarded a PhD from Coventry University, UK, in 1997. He has a broad interest in MEMS and nanotechnology and published over 200 peer reviewed publications.

H. Vogt received the Diploma degree in electrical engineering and the Dr.-Ing. degree from the University of Dortmund, Dortmund, Germany, in 1981 and 1986, respectively. Since 1986, he has been with the Fraunhofer Institute of Microelectronic Circuits and Systems (IMS), Duisburg, Germany, where he is currently the Deputy Director and the Head of the department on technology R&D. He is heading a facility for the co-integration of MOEMS and CMOS, and directs groups on development for MEMS and infrared imaging devices. He is also a Professor with the Department of Electrical Engineering and Information Technology, University of Duisburg-Essen, Duisburg.